



Mavens

What is POM.xml? (Project Object Model)

The **POM.xml** is an **XML configuration file used by Maven** that defines how a Java project should be **built, tested, packaged, managed, and deployed**.

It acts as the **blueprint** or **heart** of a Maven project.

Why POM.xml is Important?

- It tells Maven **what the project is, what it depends on, how to build it, and what plugins/tools** should be used.
 - It provides a **standardized and automated build process** across all environments (dev, test, prod).
 - It ensures **repeatability**—anyone can clone the project and build it with the same configuration.
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Key responsibilities of POM.xml

- Project Metadata(group ID, artifact ID(name of project), version etc)
 - Dependency Management(all the details related to dependencies so that later Maven can install these dependencies automatically without installing dependencies one by one manually)
 - Build & plugin configuration(specify different plugins to inform maven how that particular part would be handled for ex java plugin to tell details about java version)
 - Repository configuration
 - Profiles for Environments
 - Project Inheritance and multi module support
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Maven Lifecycle

A **lifecycle** is a sequence of steps (called **phases**) that Maven executes in a defined order to build and manage a project.

It ensures a **standard, automated, repeatable build process** across all environments.

Maven has **3 major lifecycles**:

1. **clean** → Prepare workspace
2. **default (build lifecycle)** → Build and package the project
3. **site** → Generate documentation



What is Dependency Management?

Dependency Management in Maven is the mechanism that allows you to **declare, control, download, and version-manage** all external libraries your project needs.

Maven automatically fetches these dependencies from repositories and ensures consistent builds across environments



Key Features of Maven Dependency Management

- Automatic Dependency Resolution
- Centralized Version Control
- Dependency Scopes
- Dependency Management Section (Parent POM)
- Repository Integration

